

DRONE SWARM: A Kingmaker Of New Technology?

Researchers are developing cyber solutions to defeat enemy drone swarm attacks. These solutions contain a multi-spectral sensor system that detects and characterises drone swarms. Using cyber electromagnetic activity, swarms are disabled so that life threat to native war fighters and citizens is avoided



Swarm of security drones with surveillance camera flying in the sky



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A drone swarm is a mass of tiny drones flying like a flock of birds that can perform coordinated tasks, controlled by an expert operator. A single operator from a portable ground unit can control up to fifty drones that can fly up to a hundred kilometres and carry payloads of one kilogram each, spending about an hour on the target.

Flocking drones are slowly becoming a major technological issue. Flocking behaviour is off course, and is common in the natural world, particularly among birds, fish and insects. It is now being observed in the technological world as well via drone swarms. We have seen that a large number of organisms can move collectively, maintaining direction and not hitting obstacles or each other.

In February 2018, 1218 drones formed Olympic rings in the sky at the opening ceremony of Winter Olympics in South Korea.

China has already tested a swarm of a thousand drones. Size of a swarm mainly depends on its mission or application.

Applications of a drone swarm are many, some of which include:

- Stage entertainment
- Hobbyist war games
- Animal herding on farms
- Search and rescue
- Spot-spraying during fire outbreaks, as in wildfires and the like
- Hunting thieves and spraying them with non-erasable ink
- Wi-Fi coverage
- Delivery of goods
- National and homeland security

Drone swarms have bigger applications in space exploration and can make space exploration faster, better and cheaper.

Conventional concerts can be sometimes boring, as these involve artists singing songs, with a bunch of spotlights